

TAGDALEN, Norway

WMO: 012210

Lat: 63.056N

Lon: 9.072E

Elev: 415

StdP: 96.44

Time Zone: 1.00 (EUC)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -14.7	(c) -12.4	(d) -19.6	(e) 0.7	(f) -12.2	(g) -17.8	(h) 0.8	(i) -10.7	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 8.2	(c) 24.2	(d) 15.4	(e) 21.9	(f) 14.3	(g) 20.0	(h) 13.2	(i) 16.2	(j) 22.9	(k) 15.0	(l) 20.7	(m) 13.9	(n) 18.9	(o) N/A	(p) N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
13.3	10.0	18.3	12.3	9.3	17.1	11.2	8.7	16.4	46.8	23.3	43.2	20.7	40.2	18.9	21.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) N/A	(b) N/A	(c) N/A	(e) -16.4	(f) 27.2	(g) 3.3	(h) 2.9	(i) -18.8	(j) 29.3	(k) -20.7	(l) 30.9	(m) -22.6	(n) 32.5	(o) -25.0	(p) 34.6		
(4) N/A	(5) N/A	(6) N/A	(7) -17.0	(8) 18.0	(9) 3.1	(10) 1.7	(11) -19.2	(12) 19.3	(13) -21.0	(14) 20.3	(15) -22.7	(16) 21.3	(17) -25.0	(18) 22.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	4.3	-3.4	-2.5	-0.9	3.1	7.2	10.1	13.5	12.8	9.5	4.4	0.0	-2.4
	DBStd	7.12	4.32	4.27	3.60	3.63	4.52	3.63	3.96	3.10	3.27	3.68	4.37	4.55
	HDD10.0	2395	415	351	337	209	114	42	9	8	49	176	300	383
	HDD18.3	5128	673	584	596	458	347	247	160	174	266	431	550	641
	CDD10.0	322	0	0	0	2	26	46	117	94	33	3	0	0
	CDD18.3	11	0	0	0	0	0	1	9	1	0	0	0	0
	CDH23.3	106	0	0	0	0	3	9	83	10	0	0	0	0
	CDH26.7	14	0	0	0	0	0	0	14	0	0	0	0	0
Wind	WSAvg	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Precipitation	PrecAvg	1495	146	138	142	79	78	108	101	131	146	137	133	155
	PrecMax	1891	407	312	355	220	164	208	169	294	412	271	355	343
	PrecMin	1004	4	3	35	17	45	42	30	45	51	27	4	36
	PrecStd	217	93	72	86	52	27	43	38	60	87	51	98	71
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	6.3	7.8	9.2	17.0	23.2	24.8	28.4	24.7	20.6	16.0	10.8	7.5
		MCWB	1.8	3.2	3.5	9.6	13.7	14.1	18.6	15.9	13.4	9.9	6.8	3.5
	2%	DB	4.5	5.5	6.9	13.7	20.6	21.2	25.9	22.0	18.3	13.8	8.4	6.0
		MCWB	1.3	1.4	2.4	6.7	12.0	13.5	16.7	14.6	12.2	8.4	4.9	2.6
	5%	DB	3.2	4.1	5.3	10.8	18.1	18.6	23.1	20.2	16.4	11.3	7.0	4.7
		MCWB	1.2	0.7	1.7	5.3	10.8	12.2	15.0	13.5	11.2	6.9	4.2	1.7
	10%	DB	2.1	2.9	4.1	8.6	15.2	16.6	20.5	18.3	14.6	9.4	5.5	3.4
		MCWB	0.5	0.4	1.4	3.9	9.3	11.1	14.2	12.9	10.3	5.7	2.9	1.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.3	3.7	4.6	10.0	14.4	15.4	19.6	16.7	14.4	11.6	7.5	4.5
		MCDB	4.8	6.1	6.6	16.5	22.0	21.6	28.1	22.7	18.9	14.9	10.2	6.2
	2%	WB	2.4	2.4	3.6	7.2	12.6	14.0	17.1	15.4	13.0	8.5	5.7	3.3
		MCDB	3.4	4.3	5.3	12.8	19.5	20.9	24.6	20.7	16.8	13.0	7.7	5.4
	5%	WB	1.5	1.6	2.7	5.6	11.0	12.6	15.8	14.4	11.6	7.2	4.3	2.2
		MCDB	2.7	3.3	4.7	10.1	17.2	18.3	22.7	19.1	15.3	10.7	6.5	4.1
	10%	WB	0.5	0.7	1.6	4.4	9.5	11.3	14.4	13.4	10.6	6.0	3.2	1.1
		MCDB	2.2	2.4	3.8	8.0	15.2	16.0	19.8	17.2	14.1	9.0	5.2	2.9
Mean Daily Temperature Range	5% DB	MDBR	4.8	5.2	6.4	7.4	8.4	8.0	8.2	7.8	7.0	5.9	4.8	4.7
		MCDBR	5.2	5.9	7.6	10.7	12.9	12.1	12.4	11.0	9.7	8.6	6.4	5.8
		MCWBR	3.6	3.8	4.8	5.8	6.8	6.7	6.3	5.8	5.6	5.3	4.3	4.3
	5% WB	MCDBR	4.4	5.1	6.2	9.7	12.4	11.5	12.0	10.1	9.1	7.8	5.6	5.7
		MCWBR	3.5	3.7	4.5	5.7	7.0	6.9	6.4	5.7	5.7	5.1	4.2	4.6
Clear-Sky Solar Irradiance	taub	0.223	0.280	0.312	0.340	0.342	0.343	0.352	0.340	0.318	0.295	0.232	0.364	
	taud	2.026	2.198	2.235	2.296	2.369	2.443	2.450	2.482	2.530	2.418	2.092	2.671	
	Ebn at Noon	407	658	782	841	873	879	859	839	792	654	389	235	
	Edh at Noon	31	61	88	103	104	99	96	86	67	50	28	19	
All-Sky Solar Radiation	RadAvg	0.17	0.72	2.05	3.83	4.91	5.03	4.64	3.59	2.18	0.97	0.26	0.09	
	RadStd	0.03	0.12	0.26	0.26	0.28	0.41	0.40	0.34	0.23	0.12	0.05	0.01	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (39 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	-143	-189	+67	N/A

Nomenclature: See separate page